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Diagnostics of atmospheric-pressure pulsed-dc discharge with metal and liquid anodes by multiple laser-aided methods

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Abstract

The density and temperature of electrons and key heavy particles were measured in an atmospheric-pressure pulsed-dc helium discharge plasma with a nitrogen molecular impurity generated using system with a liquid or metal anode and a metal cathode. To obtain these parameters, we conducted experiments using several laser-aided methods: Thomson scattering spectroscopy to obtain the spatial profiles of electron density and temperature, Raman scattering spectroscopy to obtain the neutral molecular nitrogen rotational temperature, phase-modulated dispersion interferometry to determine the temporal variation of the electron density, and time-resolved laser absorption spectroscopy to analyze the temporal variation of the helium metastable atom density. The electron density and temperature measured by Thomson scattering varied from 2.4×10^{14} cm^{-3} and 1.8 eV at the center of the discharge to 0.8×10^{14} cm^{-3} and 1.5 eV near the outer edge of the plasma in the case of the metal anode, respectively. The electron density obtained with the liquid anode was approximately 20% smaller than that obtained with the metal anode, while the electron temperature was not significantly affected by the anode material. The molecular nitrogen rotational temperatures were 1200 K with the metal anode and 1650 K with the liquid anode at the outer edge of the plasma column. The density of helium metastable atoms decreased by a factor of two when using the liquid anode.

Keywords: atmospheric-pressure plasma, laser diagnostics, plasma in contact with liquid, electron density, electron temperature, excited species density

S Online supplementary data available from stacks.iop.org/PSST/25/045004/mmedia

(Some figures may appear in colour only in the online journal)

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1. Introduction

Small-scale low-temperature plasmas generated at and near atmospheric pressure, typically categorized into microplasmas, have been intensively developed for various industrial applications [1–4]. In addition to atmospheric-pressure plasmas in gases, in recent years, plasmas in and/or in contact with liquids have become an emerging research field in low-temperature plasma technology [5–8]. The electric discharges with a liquid electrode have been in a research area of electrolysis since the 19th century [9, 10] and have been applied to the analytical chemistry of liquids [11, 12]. Recent studies on low-temperature plasmas in and/or in contact with liquids are highly expected to be important for various applications such as water purification [13, 14] and nanomaterial synthesis [15–18], and to be critical for biomedical applications [19–21].

To resolve the scientific and technological challenges of atmospheric-pressure plasmas, it is indispensable to know the fundamental parameters of the plasma source and understand the effects of liquids on the parameters. For this purpose, in this study, we measured the electron density and electron temperature in an atmospheric-pressure discharge plasma. The rotational temperature of ambient molecular gas and the temporal variation of excited atom density were also analyzed in order to support the investigation. These parameters have been widely utilized as basic and/or primary output parameters in simulation studies of atmospheric-pressure plasmas [22, 23]. The investigated plasma source was a pulsed-dc discharge that can generate plasmas both on metal and on liquid anodes in an atmospheric-pressure open ambient. The plasma was generated in a miniature helium (He) gas flow surrounded by a nitrogen (N_2) shielding flow to prevent any effects of ambient air, such as humidity, on the plasma parameters [24]. The liquid electrode was water and its conductivity was adjusted by adding sodium chloride (NaCl).

The laser diagnostic methods used to obtain the electron density (n_e) were laser Thomson scattering spectroscopy (LTS) [25] and phase-modulated dispersion interferometry (PMDI) [26–28]. Various n_e measurements [29–31] have been applied to atmospheric-pressure plasmas; however, the n_e measurement using the two laser diagnostics applied to an atmospheric-pressure plasma source is the first attempt to the best of our knowledge. LTS was also used to measure the electron temperature (T_e), and the Raman scattering spectra were recorded at the same time since the He flow contained a N_2 impurity entrained from the shielding flow. The rotational Raman spectra provided information on the rotational temperature (T_{rot}) of the N_2 impurity heated by the plasma generation [32, 33]. Temporal evolutions of the He metastable atom (He*) density (n_{He^*}) were diagnosed in relative values by laser absorption spectroscopy (LAS) [34, 35] as a means of supporting the n_e diagnostics, similarly to in the previous studies [36, 37].

The combined application of sophisticated diagnostic methods to a single plasma source has been performed for fusion plasmas [38, 39] and conventional low-pressure low-temperature plasmas [40, 41] for decades, and for some atmospheric-pressure plasmas in recent years [37, 42–44]. It enables us not only to obtain various plasma parameters but also to reveal characteristic features which that cannot be revealed by a single diagnostic method, making a greater contribution to subsequent simulation and application studies.

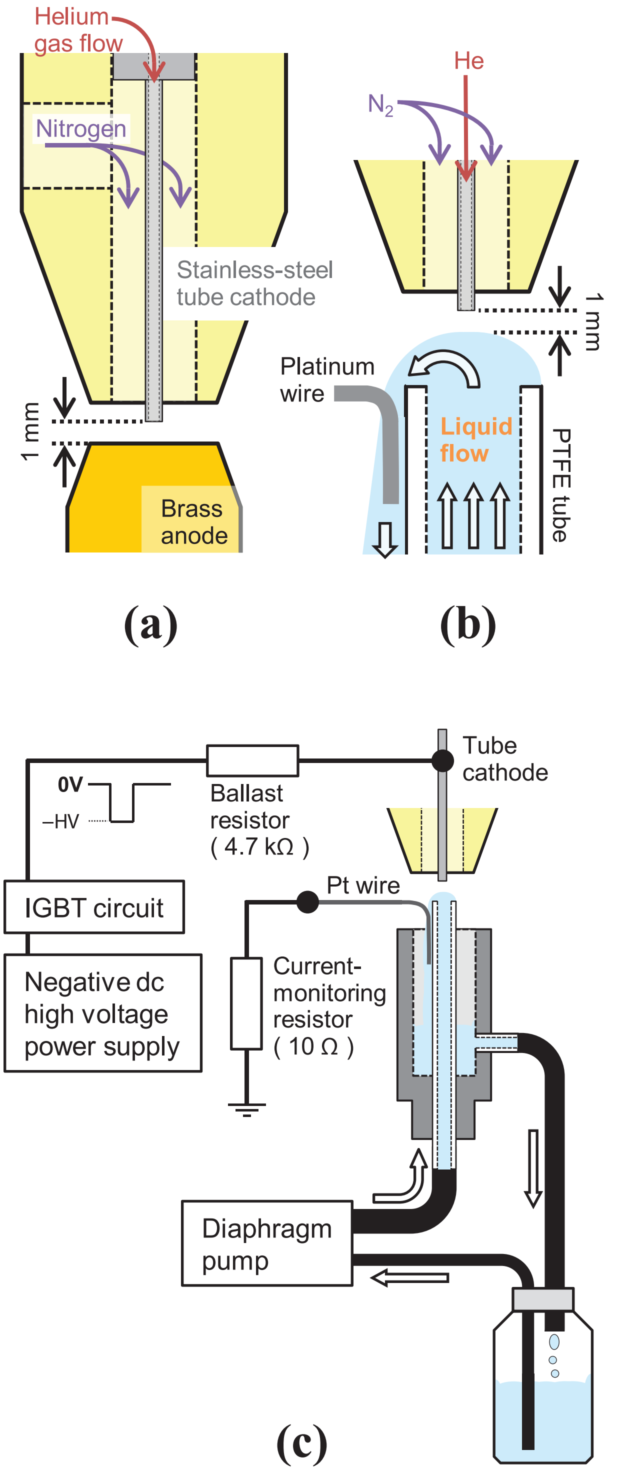


Figure 1. Schematics of atmospheric-pressure plasma generation using pulsed-dc discharge system in this study. (a) Electrode and gas flow structures composed of a stainless-steel tube cathode and brass rod anode, and He core and N_2 shielding gas flows. (b) Continuously flowing liquid anode and platinum (Pt) wire with the tube cathode and gas flows. (c) Electric circuit applying pulsed-dc high voltage to the tube cathode, current-monitoring resistor, and water flowing system.

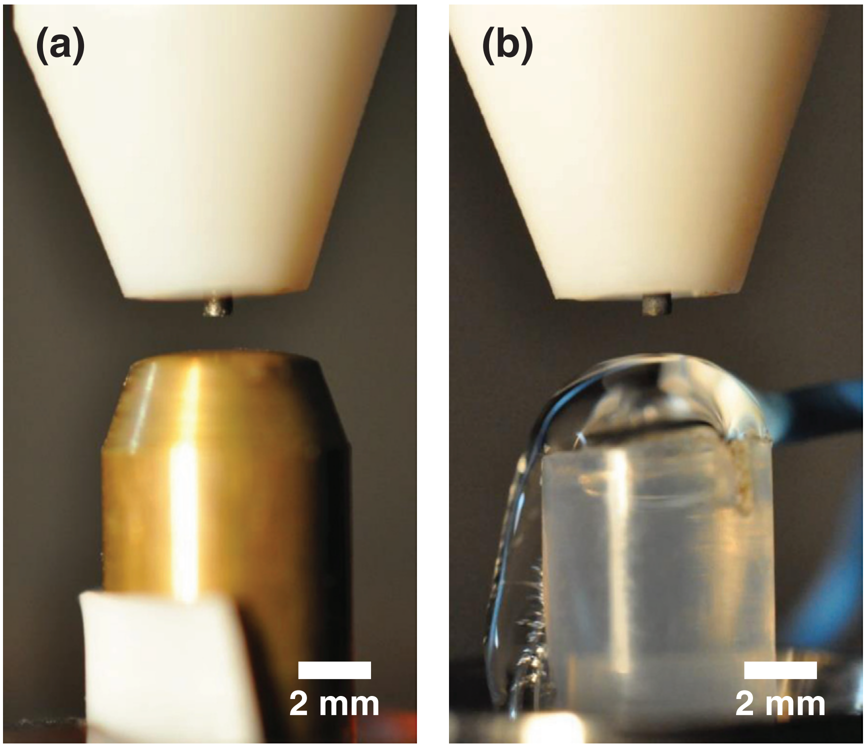


Figure 2. Photographs of plasma source with (a) metal and (b) flowing liquid anodes used in this study. The wire with the blue insulation appearing on the right-hand side in (b) is the Pt wire used to form a discharge-current flow path in the downstream water flow in the lower left of the photograph.

The paper is organized as follows. In section 2, we describe the experimental setups and procedures including the atmospheric-pressure plasma source, the observation of the plasma emission, and the laser diagnostic methods. The experimental results of the observation and laser diagnostics are presented in section 3, which are separated into each diagnostic method. In section 4, we discuss and summarize the plasma parameters measured in the cases of metal and liquid anodes. Finally, we conclude this study in section 5.

2. Plasma source and diagnostic methods

2.1. Atmospheric-pressure pulsed-dc discharge plasma source

The tested plasma source was a pulsed-dc discharge in a miniature He gas flow. As shown in figure 1, the He flow was ejected from a stainless-steel tube cathode with 0.5 mm inner and 0.8 mm outer diameters and its flow rate was 200 sccm, corresponding to a flow speed at 20 m s^{-1}. To form a shield between the core He flow and the ambient air, we used an additional N_2 gas flow from a plastic tube with an inner diameter of 4 mm surrounding the tube cathode with a flow rate of 500 sccm (flow speed: 0.7 m s^{-1}). The distances traveled by these flows during one discharge pulse (1 ms in this study) were 20 mm for the He core and 0.7 mm for the N_2 shielding flow; therefore, the N_2 shielding flow acted as a stationary ambient during the pulse. As the metal anode, we used a brass rod with a polished surface that was placed 1 mm from the edge of the tube cathode (figure 1(a)). Figure 2(a) shows a photograph of the plasma electrode structure with the metal anode.

In the case of the liquid anode, we prepared a continuously flowing liquid electrode (figure 1(b)). The flow system can maintain the same liquid conditions for long-time operation, which is necessary for the laser diagnostics. This type of discharge using a flowing-liquid electrode is employed in research on analytical chemistry [11, 12] and recently in research on nanomaterial synthesis [45]. The liquid used in this study was water containing dissolved NaCl with a fixed conductivity of 15 mS cm^{-1}, monitored by a conductivity meter (Hanna Instruments, DiST6). The NaCl solution was flowed from a plastic tube with 4.35 mm inner diameter at a flow rate of 45 ml min^{-1} using a cyclic system that consisted of a bottle, a diaphragm pump, and an electrode as shown in figure 1(c). The liquid anode was grounded via a platinum (Pt) wire immersed in the flowing liquid immediately downstream from the edge of the plastic tube. So that the discharge conditions were the same as those for the metal anode, the tube cathode was set on the top of the hemispherical water flow and the distance of the gap between the tube edge and the top of the water surface was set to 1 mm as shown in the photograph in figure 2(b).

Figure 1(c) shows the electrical system used for plasma generation. A negative dc high voltage from a power supply (Matsusada Precision, HAR-3R733) was modulated to a square pulse by a half-bridge insulated gate bipolar transistor (IGBT) module (Dynex, DIM100PHM33-F), and the negative voltage pulse was applied to the tube cathode through a ballast resistor of 4.7 kΩ. For both the metal and liquid anodes, a current-monitoring resistor of 10 Ω was connected between the anode and the ground. The pulsed applied voltage had a

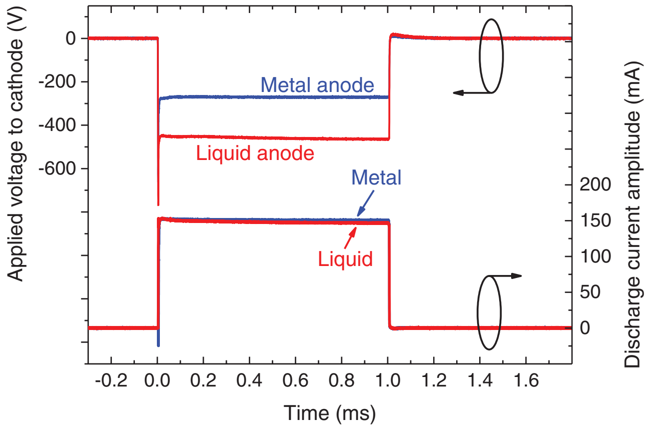


Figure 3. Measured waveforms of applied voltage to tube cathode and discharge current amplitude monitored between anode and ground for metal (blue lines) and flowing liquid (red lines) anodes.

frequency of 100 Hz and a duty ratio of 10%, corresponding to a pulse width of 1 ms. We monitored the applied voltage to the cathode and the discharge current, and maintained the current amplitude at 150 mA for both the metal and liquid anodes. Waveforms of the applied voltage and the discharge current amplitude are plotted in figure 3. Since there was an additional resistance between the top surface of the weter and the Pt wire, the amplitude of the applied voltage to flow the same current was higher for the liquid anode than for the metal anode.

2.2. Observation of plasma emission

To investigate the plasma structures and their temporal stability, we took visible-emission photographs and videos using a commercial digital camera (Nikon, D5000) with a macro lens (Nikon, AF-S Micro NIKKOR 40 mm 1:2.8G). The shutter speed used for plasma imaging was 0.01 s. The video taken by the digital camera is included in the supplemental data.

In addition to the digital camera, a high-speed video camera (Photron, FASTCOM-SA3) connected to a microscope lens (Leica, Z16APO) was used to observe the temporal variation of the plasma emission structures during one discharge pulse. The plasma was observed at a temporal resolution of 50 000 fps (frames per second), which was sufficiently high to resolve the pulsed-dc discharge with a pulse width of 1 ms. High-speed video observation has already been applied to atmospheric-pressure plasmas on liquid electrodes, for example, to investigate the self-organized pattern formed on the liquid surface [24, 46].

2.3. Laser Thomson and Raman scattering spectroscopy

Laser Thomson scattering (LTS) can give local values of n_e and T_e with high spatial and temporal resolutions, and it has been applied to investigate n_e and T_e for plasmas generated in high-pressure atmospheres including laser-produced plasmas [47–49], arc discharges [50–52], and high-pressure low-temperature plasmas [33, 53–55]. The Thomson scattering is the scattering of radiation (induced laser beam) by free charged particles (electrons in plasma), and it includes information of n_e (integrated area of the scattering spectrum) and T_e (the spectrum profile) at the scattering position. In addition, when LTS is applied to plasmas including molecular gases, rotational Raman scattering spectra are also recorded superimposed on the Thomson scattering. While the superimposed Raman spectra make n_e measurement difficult in some cases, they are generally able to provide information on the molecular density and rotational temperature T_{rot}. Recorded rotational Raman spectra are sometimes used to estimate the gas temperature. However, according to [33], the gas temperature at the center of a plasma jet measured by Rayleigh scattering (600 K) is higher than T_{rot} for an air impurity at the same position (390 K). The reason for the difference is the vibrational excitation of N_2 molecules inside the plasma suggested by Carbone and Nijdam [56]. Therefore, it should be noted that the gas temperature are probably different from T_{rot} for the N_2 impurity inside the plasma column also in our study.

The laser Thomson and Raman scattering measurements in this study were performed using the second harmonic of a Nd:YAG laser beam (Continuum, Surelite III) with a 532 nm wavelength and a 10 ns pulse as the input laser. In high-pressure plasma diagnostics, care must be taken to avoid perturbation of the plasma parameters by the laser beam itself [57, 58], and we set the input laser energy to less than 10 mJ, for which we confirmed that no plasma perturbation occurred in the setup. The laser beam was focused onto the center of the discharge gap with a spot size of 120 μm in full width at half maximum (FWHM). Scattered light was collected with an angle of 90° to the laser axis and focused onto the entrance slit of a triple-grating spectrometer using two of achromatic lenses with 220 mm focal length and 45 mm effective diameter. The Thomson and Raman scattering spectra were recorded by an intensified charge-coupled device (ICCD) camera (Princeton Instruments, PI-MAX UNIGEN II) with over 2000 accumulations for one measurement to obtain a sufficient signal-to-noise (S/N) ratio. A spectral resolution of the system was 0.34 nm. The measurement apparatus and analysis procedures used in this study are also described in [59].

2.4. Phase-modulated dispersion interferometry

Dispersion interferometry (DI) is a tool used to determine the value of n_e in fusion plasmas [60, 61], and in a low-temperature plasma (low-pressure xenon discharge tube) [62]. The interferometry detects the change of refractive index due to the generation/extinction of electrons inside the plasma as the phase shift of the probing laser beam, and the line-integrated n_e along the laser path can be calculated from the refractive-index variation. Applying phase modulation to the probing laser beam increased accuracy of the signal processing by avoiding the effect of the variation of the detected laser intensity on the phase-shift measurement. Such phase-modulated dispersion interferometry (PMDI) was first reported in 2010 [26–28] and has been used in the large helical device (LHD at National Institute for Fusion Science, Japan) [63, 64]. In 2014, it was found that PMDI also makes it possible to determine n_e

in high-pressure plasmas with a high resolution of n_e since it automatically cancels the phase shift induced by the variation of the gas number density in and around the plasma, which is significant in high-pressure plasmas [65].

In this study, we used the same PMDI system as in our previous study [65], which is briefly described as follows. The probing laser was a CO_2 laser (MPB Technology, GN-802-GES) with a wavelength of 10.6 μm, which was passed through $AgGaSe_2$ nonlinear crystal (Crystech) to generate a second-harmonic beam at 5.3 μm. The beam containing fundamental and harmonic components was focused onto the center of the discharge gap by a pair of ZnSe lenses with 100 mm focal length. The diameter of the probing beam at the plasma was approximately 240 μm. The harmonic beam was phase modulated at 50 kHz in a photoelastic modulator (HIND Instruments, II/ZS50). Then, the beam with the fundamental and harmonic components passed through a second nonlinear crystal, and the interference beat signal between the second-harmonic beams generated before and after the plasma was recorded by a detector (Vigo System, PVI-3TE) and sent to a signal processing system having two lock-in amplifiers (NF Corporation, LI5640). In the lock-in amplifiers, the signal amplitude at the frequencies of phase modulation ($\omega_{PM} = 50$ kHz) and its second harmonic ($2\omega_{PM} = 100$ kHz) in the total detected signal were calculated continuously. To improve the S/N ratio in the measurement, we averaged the output signal amplitudes from the lock-in amplifiers over 4000 times, where the output signals were synchronized with the discharge pulse in a digital oscilloscope (Yokogawa Electric, DL750).

From the arctangent of the amplitude ratio between the frequency elements of the fundamental $I_{\omega_{PM}}$ and second harmonic $I_{2\omega_{PM}}$ of the phase modulation frequency, we can calculate n_e using the following relationship [65].

$$\arctan\left(\frac{I_{\omega_{PM}}}{I_{2\omega_{PM}}}\right) = \frac{3e^2\lambda}{8\pi c_0^2 m_e \varepsilon_0}\int n_e \mathrm{d}l + \frac{12\pi AB}{\lambda^3 n_{g0}}\int \Delta n_g \mathrm{d}l \tag{1}$$

Here, e is the charge of an electron, λ is the probing laser wavelength ($\lambda = 10.6$ μm for the PMDI in this study), c_0 is the light speed in vacuum, m_e is the mass of an electron, A and B are specific constants that depend on the gas species [66], n_{g0} is the gas number density n_g at STP (standard temperature and pressure), and $\Delta n_g = n_g(t) - n_g$ $(t = 0)$ is the variation of n_g due to the generation of plasma. In equation (1), there is still a term of n_g which can be varied by the gas heating. However, since the n_g term is related only to the dispersive component in the refractive index of gas which is ignored in almost every case of scientific studies, its amplitude is significantly smaller than that in interferometry without the 'dispersion' system.

2.5. Laser absorption spectroscopy

Laser absorption spectroscopy (LAS) is a broadly utilized diagnostic method for high-pressure plasmas, whose main measurement target is the density of excited species and radicals. LAS studies, including those with absorption enhancement methods such as cavity ring-down spectroscopy, on high-pressure small-scale plasmas have so far measured He, argon, xenon, and krypton metastable atoms [34–36, 67, 68], N_2 metastable molecules [69], and OH radicals [70].

We used LAS in this study to analyze the temporal change of relative density of the excited He atoms in the 2^3S_1 metastable state (He*). A diode laser beam (Sacher Lasertechnik, TEC500) entered the center of the discharge gap, where the wavelength of the beam was $\lambda = 1083.035$ nm, which is the peak absorption wavelength of the $2^3S_1 - 2^3P_2$ transition. The beam diameter was approximately 150 μm in the plasma column. The transmitted probing laser beam was detected by a photodetector (Thorlabs, PDA10CF) with a response time of approximately 10 ns. The laser beam wavelength was monitored by a wavelength meter (Coherent, WaveMate Deluxe) and controlled by a piezo actuator inside the external cavity of the laser source. To improve the S/N ratio of the measured data, we averaged the measured temporal evolutions over 3000 times, where the photodetector signal was synchronized with the discharge pulse in a digital oscilloscope (LeCroy, WaveRunner 610Zi). The electric noise and plasma emission in the detected LAS signals were eliminated by subtracting the signal measured without the probing laser beam in the same discharge conditions.

The He* density n_{He^*} is related to the transmittance of the laser beam I/I_0 using as the following equation [36, 44].

$$-\ln(I/I_0) = \int k \mathrm{d}l \cong C\int n_{He^*}\mathrm{d}l \tag{2}$$

Here, I and I_0 are the transmitted and incident probing laser-beam intensities, respectively, $-\ln(I/I_0)$ is the absorbance, k is the absorption coefficient of the He* atom at the fixed wavelength, and C is the constant. The C value can be derived from Doppler and collisional broadenings in the absorption profile considering radial distributions of the gas temperature and gas composition [29, 44]. However, since the radial distributions of these parameters for the discharge in this study have not been clarified yet, we show the LAS results in the absorbance $-\ln(I/I_0)$, which represents relative n_{He^*} in a discharge condition, not in absolute n_{He^*}.

3. Results

3.1. Imaging of discharge structures

The photographs in figure 4 show the plasma structures in the cases of the metal (figure 4(a)) and liquid (figure 4(b)) anodes. Comparing the photographs, the plasma structures are similar near the tube cathode, whereas the structures are clearly different between the center of the discharge gap and the metal and liquid anodes. In the case of the metal anode, the discharge diameter slightly decreases toward the anode and there are anode spot emissions on the surface, which are typically observed in atmospheric-pressure glow discharges as reported in [71–73]. In contrast, in the case of the liquid anode, the discharge diffuses toward the anode and no anode spot was observed on the liquid surface, similarly to in previous studies of plasmas on liquid electrodes [11, 24, 74]. In addition to

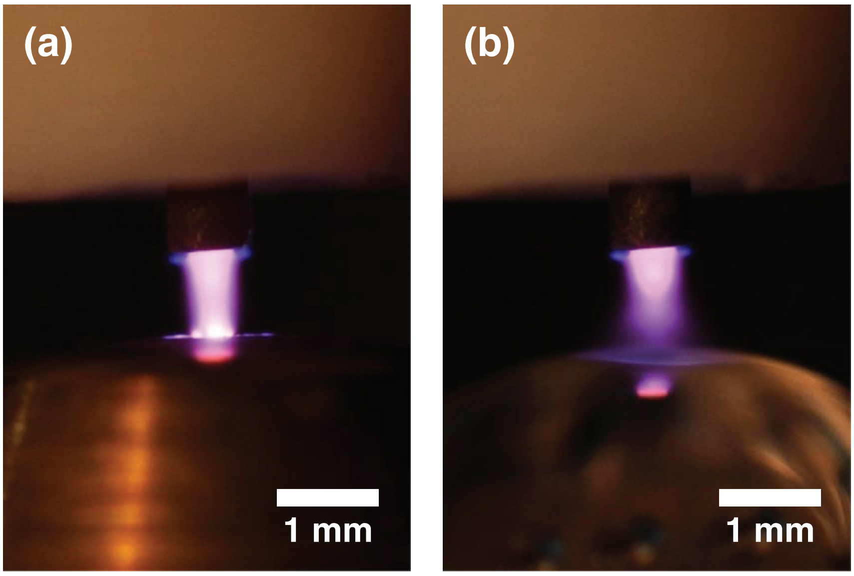


Figure 4. Photographs of plasma emissions taken with (a) metal and (b) flowing liquid anodes. The shutter speed for the two photographs is 0.01 s. The emission spot appearing below the plasma column is an image of the cathode emission reflected on the anode surface.

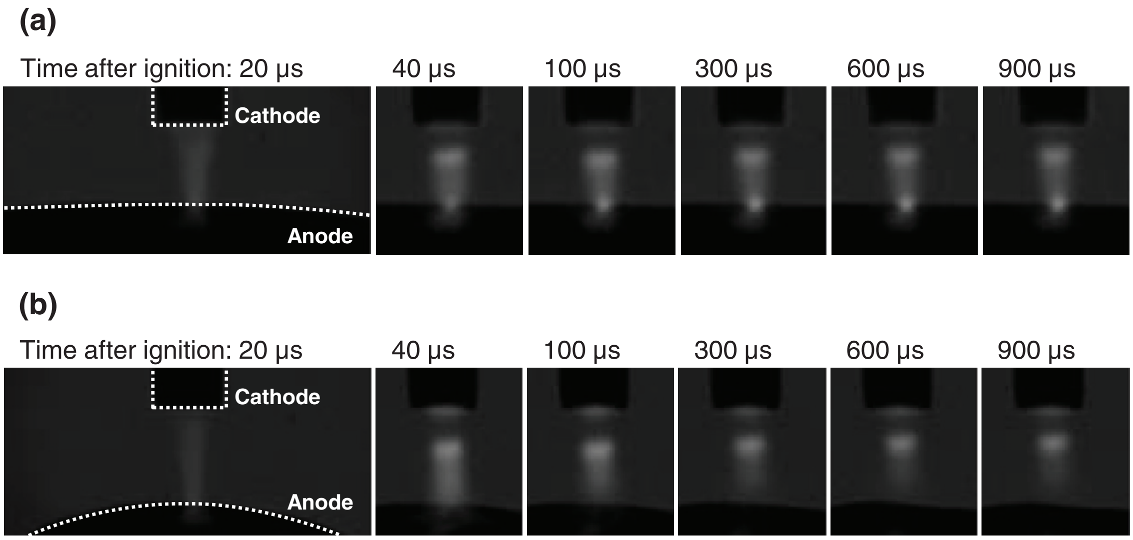


Figure 5. High-speed video camera images of plasma emissions taken with (a) metal and (b) flowing liquid anodes. The frame speed of the high-speed camera is 50 000 fps, which is sufficient to resolve one pulse of the pulsed-dc discharge with 1 ms duration. The time above each image indicates the time after the discharge ignition when it was taken.

the photographs in figure 4, video images taken by the same camera are available online as supplementary data (stacks.iop.org/PSST/25/045004/mmedia, see the file 'Movie1.mp4').

Temporally resolved images are shown in figure 5. For the case of the metal anode (figure 5(a)), the plasma emission structures observed from 40 μs after the discharge ignition are identical and the cathode dark region and the anode spot can be seen. The cathode dark region was also observed in the case of the liquid anode (figure 5(b)); however, the lower part of the discharge emission, particularly near the liquid anode, gradually diffuses and its intensity decreases toward the pulse-off timing. The discharge gap between the cathode edge and water surface slightly varied during the discharge. Also, it was confirmed that the plasma structures and their temporal evolutions were identical in each discharge pulse. We did not observe any fluctuation of water surface from the high-speed camera videos in both time periods with and without the plasma. It suggests that local turbulence and/or boiling did not perturb the water surface in a scale which can be seen in the camera resolutions in time and space. The original video images are available online as supplementary data (see the file 'Movie2.mp4').

3.2. Laser Thomson and Raman scattering spectroscopy

We acquired 2D ICCD camera images of Thomson and Raman scattering signals when the probing YAG laser beam was passed through the plasma 600 μs after the discharge ignition as shown in figures 6(a) and (b). In these ICCD images, the horizontal axis is the radial position r of the plasma and the gas flow column and the vertical axis is the wavelength difference from the probing-laser wavelength $\Delta\lambda$. The dark part at the probing-laser wavelength is due to the reverse slit. Figure 6(a) shows the ICCD image taken in the case of the metal anode and figure 6(b) shows the image taken in the case of the liquid anode.

In the outer region more than 0.5 mm from the center, the recorded signals indicated by dotted arrows were all assigned to the rotational Raman scattering spectra of the shielding N_2 gas. We measured the rotational temperature T_{rot} of N_2 in the outer edge of the plasma column as shown in figure 7. By investigating the rotational distribution of the Raman spectra for the metal (figure 7(a)) and liquid (figure 7(b)) anodes, T_{rot} were evaluated approximately to be 1200 K for the metal anode and 1650 K for the liquid anode. The errors of the T_{rot} measurement was approximately ± 150 K estimated by the fitting with various T_{rot} values.

Near the center ($r < 0.5$ mm), since there is the core He gas with no rotational Raman scattering, the detected signal indicated by the solid arrow is the spatial profile of Thomson scattering, which provides information on n_e and T_e. Figures 8(a) and (b) show the LTS spectra containing small Raman scattering signals of the N_2 impurity, plotted by integrating over the central five pixels in the radial position, corresponding to a length of 77 μm. Before obtaining n_e and T_e, the Raman scattering signal has to be subtracted from the total measured spectra. For the fitting of the Raman scattering, we used the T_{rot} values measured in the outer edge (1200 K for the metal and 1650 K for the liquid anodes), since it was difficult to evaluate the T_{rot} values in the center due to small intensities of the Raman signal. After the fitting and the subtraction of Raman scattering, we derived n_e and T_e for each five-pixel-integrated spectrum, which was calibrated using the intensity of the Rayleigh scattering obtained without the reverse slit in the triple-grating spectrometer. The Rayleigh scattering measurement was done using a small chamber filled by pure N_2 gas with the same probing laser and scattering light collection system as the discharge measurement. For example, from the data measured at the center of the radial position shown in figure 8, n_e and T_e were calculated to be 2.4×10^{14} cm^{-3} and 1.8 eV for the metal anode (figure 8(a)) and 2.0×10^{14} cm^{-3} and 1.9 eV for the liquid anode (figure 8(b)), respectively.

Figure 9 shows radial distributions of n_e (figure 9(a)) and T_e (figure 9(b)). Since the plasma is a cylindrical glow discharge, the radial n_e distribution is plotted with the zero-order Bessel function $n_e(r) = n_{e0}\, J_0(r/R)$, where $n_e(r)$ is the radial distribution of n_e, n_{e0} is the value of n_e at the center, J_0 is the zero-order Bessel function, and R is the radial position where n_e becomes zero [75]. We assumed that n_{e0} is the measured n_e at $r = 0$ and $R = 0.5$ mm where we plotted the Bessel function curves for the metal and liquid anodes. The experimental uncertainties of the LTS are as follows. For the horizontal r axis, the uncertainty originates from averaging over 77 μm (five pixels) for each plotted point. The error of n_e was due to the statistical fluctuation in the number of detected photons and the reproducibility of the Rayleigh scattering measurement used for the calibration. The estimated n_e error was approximately 4×10^{13} cm^{-3} in the experimental setup. Also,

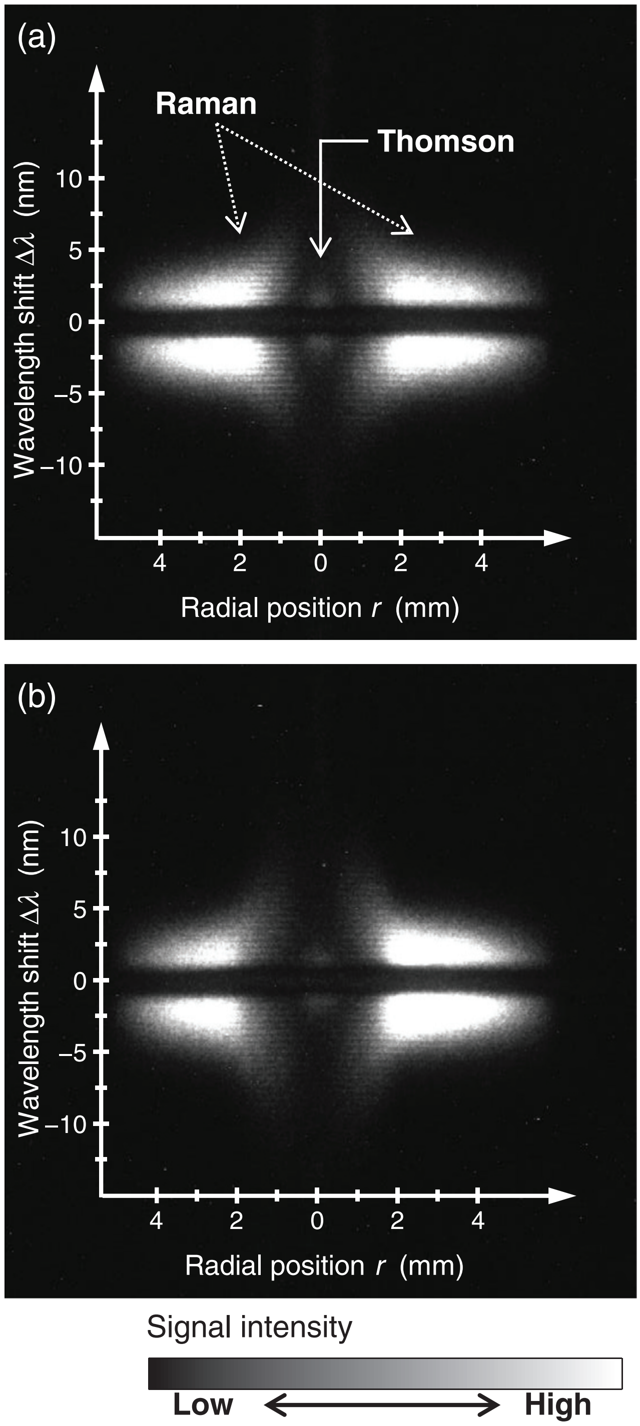


Figure 6. 2D images of laser Thomson and Raman scattering signals recorded by ICCD camera inside triple-grating spectrometer for (a) metal and (b) flowing liquid anodes. The horizontal and vertical axes show the radial distance r from the center of the plasma column and the wavelength shift $\Delta\lambda$ from the probing-laser wavelength, respectively. The scattering light near the probing-laser wavelength ($\Delta\lambda = -1.5$–1.5 nm) was eliminated by an optical slit installed in the spectrometer. The signals in the regions $r < 0.5$ mm and $r > \sim 1$ mm are the signals of Thomson scattering from electrons and rotational Raman scattering from N_2 molecules, respectively.

we estimated the T_e errors in the Gaussian fitting of scattering spectra. For example, at the center in the case of the liquid anode (data shown in figure 8(b)), the T_e error was estimated

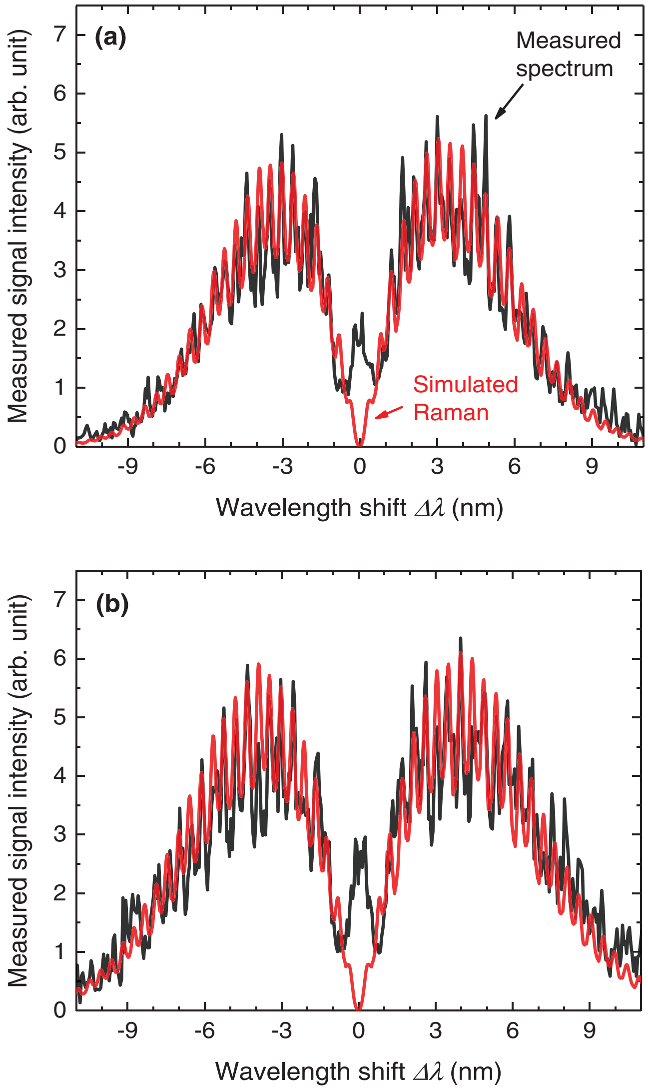


Figure 7. Rotational Raman scattering spectra of N_2 molecules measured in the outer edge of the plasma column for cases of (a) metal and (b) flowing liquid anodes. The figures show the 10-pixel integrated spectra including a range of $400 < r < 500$ μm in figure 6. Black lines are the measured spectra and red lines are the simulated Raman spectra with (a) $T_{rot} = 1200$ K and (b) $T_{rot} = 1650$ K, respectively.

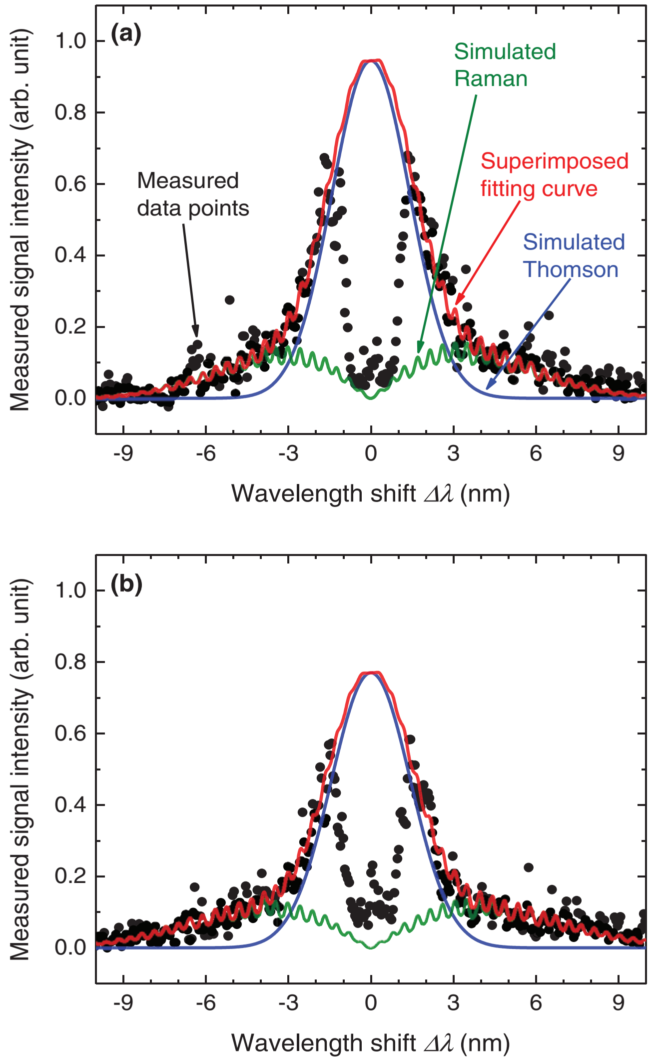
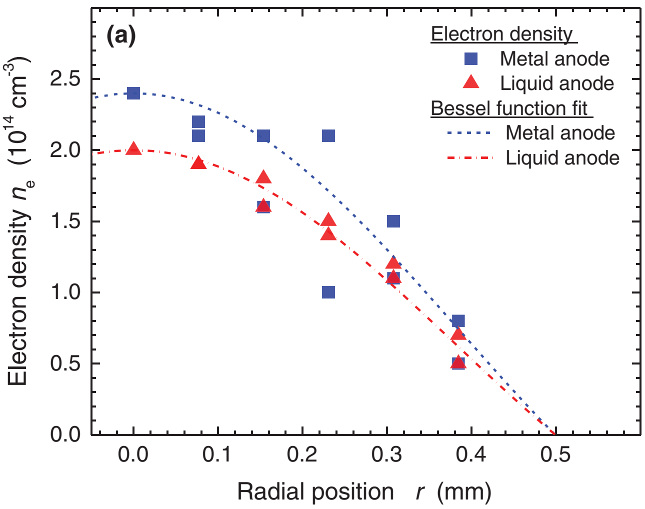


Figure 8. Measured Thomson and rotational Raman scattering signals measured at the center of the plasma column for cases of (a) metal and (b) flowing liquid anodes. The figures show the integrated spectra in the range of $r < 38$ μm in figure 6. Black dots are the detected signal intensities at each detecting element in the ICCD camera. Blue lines are the simulated Thomson scattering spectra with electron density $n_e = 2.4 \times 10^{14}$ cm^{-3} and electron temperature $T_e = 1.8$ eV for (a) and $n_e = 2.0 \times 10^{14}$ cm^{-3} and $T_e = 1.9$ eV for (b). Green lines are the simulated rotational Raman scattering spectra with rotational temperature $T_{rot} = 1200$ K for (a) and $T_{rot} = 1650$ K for (b), which were the T_{rot} measured in the outer edge, used in order to isolate the Thomson scattering from the total signal. Red lines are the best-fit curves for each measured spectrum, obtained by superimposing the simulated Thomson and Raman scattering spectra.

to be ± 0.2 eV. In the outer region, since the signal intensity of Thomson scattering became small, the T_e errors are more significant than those near the center. The analysis results for the LTS spectra suggest that at the center of the discharge gap, n_e for the liquid anode was approximately 20% smaller than that for the metal anode but with a similar spatial distribution, while no difference in T_e was clearly observed between the two cases. Such a small difference in n_e and the same T_e were acceptable considering a fact that the discharge current amplitude was limited at 150 mA in both cases.

3.3. Phase-modulated dispersion interferometry

The measurement results for the temporal evolution of $\arctan(I_{\omega_{PM}}/I_{2\omega_{PM}})$, which is directly related to n_e as indicated by equation (1), are plotted in figure 10 along with the waveform of the discharge current. The temporal variation of $\arctan(I_{\omega_{PM}}/I_{2\omega_{PM}})$ for the metal anode (blue line in figure 10) generally had similar characteristics to those in our previous study [65]: a sudden increase immediately after the ignition due to an increase in n_e, a gradual decrease due to an increase in T_g (decrease in n_g), a sudden decrease due to a decrease in n_e resulting from plasma extinction, and a gradual increase to the original value due to a decrease in T_g. From the increase in $\arctan(I_{\omega_{PM}}/I_{2\omega_{PM}})$ of 0.0037 rad at the ignition, we calculated the line-integrated n_e along the probing laser path to



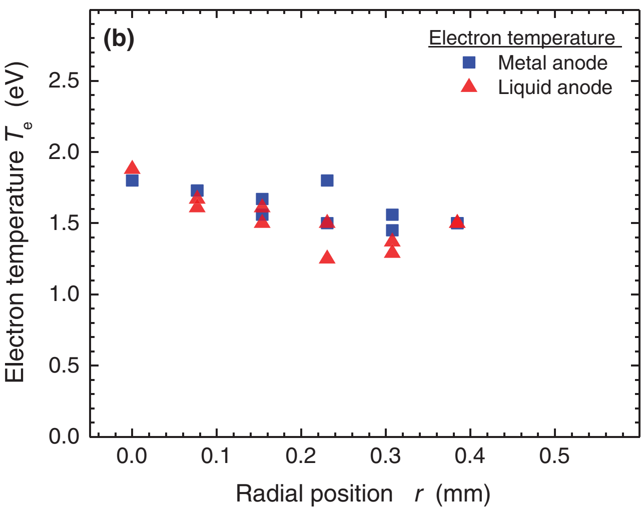


Figure 9. Radial distributions of (a) electron density n_e and (b) electron temperature T_e in the plasma with metal and flowing liquid anodes measured by LTS. The dashed and dashed-dotted lines in (a) are zero-order Bessel functions used to fit the n_e distribution calculated with the assumptions that the peak n_e value is the measured n_e at the center and n_e becomes zero at $r = 0.5$ mm. The possible errors in the vertical and horizontal directions are discussed in the manuscript.

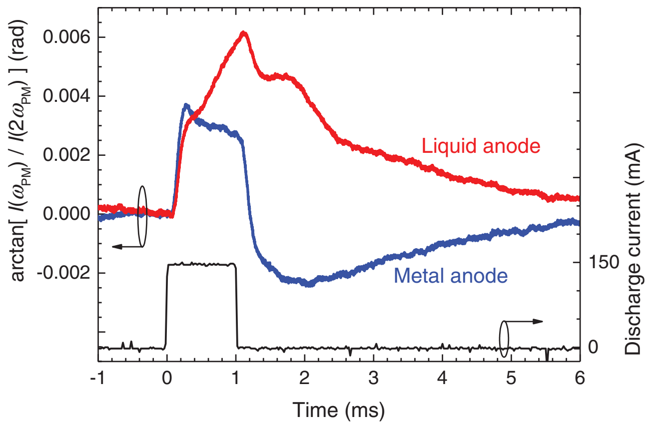


Figure 10. Temporal evolution of phase-shift signal arctan($I_{\omega_{PM}}/I_{2\omega_{PM}}$) for metal (blue line) and flowing liquid (red line) anodes measured by phase-modulated dispersion interferometry (PMDI) plotted with the discharge current waveform (black line).

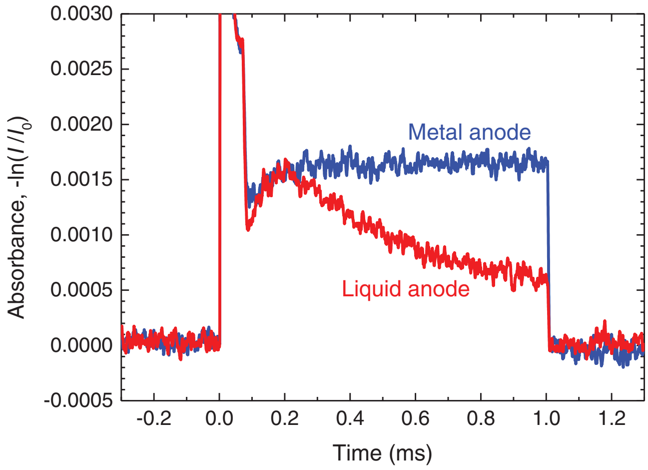


Figure 11. Temporal evolution of absorbance $-\ln(I/I_0)$ for metal (blue line) and flowing liquid (red line) anodes measured by laser absorption spectroscopy (LAS) of the He* (2^3S_1) atoms. The pulsed-dc voltage is applied from 0 to 1.0 ms.

be 8.2×10^{12} cm^{-2} from equation (1). The error of the line-integrated n_e can be estimated approximately 0.7×10^{12} cm^{-2} when we assume that only the noise in the recorded signal was a source of the error. The amplitude of the sudden decrease in arctan($I_{\omega_{PM}}/I_{2\omega_{PM}}$) immediately after the plasma extinction was similar to the amplitude of the increase immediately after in the ignition, suggesting that the value of n_e of between 0.1 and 1.0 ms after the ignition was almost constant in the case of the metal anode.

On the other hand, the temporal behaviors of the measured data in the case of the liquid anode (red line in figure 10) were different from those for the metal anode. For the n_e-related rapid change in the signal, there was a rapid increase in arctan($I_{\omega_{PM}}/I_{2\omega_{PM}}$) of approximately 0.0032 rad at the discharge ignition. At the timing of discharge extinction, the amplitude of the rapid decrease observed in 1.2 ms was approximately 0.0015 rad, which was smaller than that at the timing of ignition. This result suggests that in the case of the liquid anode, n_e decreases during the discharge pulse, and the line-integrated n_e was 7.1×10^{12} cm^{-2} for the ignition timing and 3.3×10^{12} cm^{-2} for the extinction. Also, the slowly varying component in the signal increased during the discharge pulse, had a plateau around the extinction timing and after the discharge, and then decreased toward the original position. This behavior of the slow component is opposite from the case of the metal anode, and cannot be explained by the variation in gas temperature T_g (proportional to $1/n_g$) because the arctan($I_{\omega_{PM}}/I_{2\omega_{PM}}$) is negatively proportional to T_g (see equation (1)). A decrease in T_g cannot naturally occur in an electric gas discharge, and also an increase in T_g was suggested by Raman scattering. Therefore, the signal variation is induced by another factor and it can be explained by the increase in the molecular-gas fraction inside the plasma during the discharge. This is because the A and B values in equation (1) for molecular gases (e.g. for air $A = 28.71 \times 10^{-5}$, $B = 5.67 \times 10^{-3}$) are much larger than those of He gas ($A = 3.48 \times 10^{-5}$, $B = 2.3 \times 10^{-3}$), resulting in an increase in the value of the second term on the

Table 1. Plasma parameters of the pulsed-dc discharge plasma measured by the laser-aided diagnostic methods.

Measurement method	Plasma parameters	Anode	
		Metal	Liquid
Thomson scattering (LTS)	Electron density n_e (cm^{-3})	2.4×10^{14}	2.0×10^{14}
	Electron temperature T_e (eV)	1.8	1.9
Raman scattering	N_2 rotational temperature T_{rot} (K)	1200	1650
PMDI	Electron density n_e (cm^{-3})	1.4×10^{14}	0.9×10^{14}
LAS	Temporal change of absorbance after 200 μs	Constant	50% decrease

Note: The position and time for the LTS data (n_e and T_e) are at the center of the plasma column and 600 μs after the discharge ignition, respectively. The N_2 rotational temperatures were measured at the edge of the plasma column 600 μs after the ignition. The point n_e values of the PMDI data were calculated from line-integrated data using the n_e spatial distribution measured by LTS. The average n_e value between the ignition and extinction timings is given for the liquid anode in the PMDI. The He* absorbance LAS data indicates the temporal change of relative n_{He^*} during the discharge.

right-hand side of equation (1) when the molecules are added on the probing laser axis [66]. A candidate molecule that entered on the probing laser axis (in the plasma column and its surrounding region) is water vapor, evaporated by heating of the liquid anode surface by the discharge. The increase in the water vapor ratio in a discharge space was recently measured in a similar plasma source in contact with water by laser-induced fluorescence spectroscopy, as reported by Sasaki *et al* [76] However, since we have no *A* and *B* values for water and OH radicals in a middle *IR* range, it is difficult to obtain the quantitative conclusion for which species affected the PMDI results when we used the liquid anode. The result suggested that the accumulation of *A* and *B* data in the middle IR range is important for future studies of the PMDI to analyze not only n_e accurately but also gas dynamics.

3.4. Laser absorption spectroscopy

We measured the absorbance $-\ln(I/I_0)$ of the probing laser beam at $\lambda = 1083.035$ nm as shown in figure 11. Since the temporal resolution of the LAS measurement (~10 ns) was much higher than that of the PMDI (~100 μs), a short and high-n_{He^*} excitation phase was detected in the discharge ignition from 0 to 0.1 ms. The maximum absorbance reached 0.052 in the metal and 0.043 in the liquid anode cases respectively. After the ignition phase, the discharge became stable with an absorbance of 0.0016 at 0.2 ms for both the metal (blue line) and liquid (red line) anodes. The signal fluctuation of the absorbance with an amplitude approximately 0.0002 was the main error of the LAS measurement in this study. Then, the absorbance for the metal anode was nearly constant for 0.8 ms up to the end of the discharge pulse; however, that for the liquid anode decreased toward the end of the discharge pulse. Then at the end of discharge pulse (at 1.0 ms in figure 11), it suddenly became zero since there was a N_2 impurity in the He core flow. The lifetime of He* atoms was estimated to be below 1 μs which was also reported in previous LAS studies under similar gas conditions of atmospheric-pressure He gas with N_2 impurity [37, 44].

4. Discussion

Here, to summarize and compare the measured plasma parameters, the line-integrated n_e measured by PMDI is converted to point densities at the center of the plasma column. For the conversion, we used the spatial n_e distribution measured by LTS shown in figure 9(a). By integrating the Bessel function curve used to fit the n_e distribution, the relationship between the point density at the center and the line-integrated density was determined. For the PMDI data in the case of the metal anode, assuming that n_e was constant during the discharge pulse, the point n_e at the center was calculated to be 1.4×10^{14} cm^{-3}. For the liquid anode, the point n_e at the center was calculated to be 1.2×10^{14} cm^{-3} and 0.6×10^{14} cm^{-3} at the ignition and extinction timings, respectively. And as an estimate of n_e in the middle of the discharge pulse, we used the average value of 0.9×10^{14} cm^{-3} for comparison with the LTS measurement data.

As a summary of the laser diagnostic results, the measured fundamental parameters of the plasma source for each anode are listed in table 1. Comparing n_e for the metal and liquid anodes, n_e for the liquid anode was 20% and 35% smaller than that for the metal anode according to the results of LTS and PMDI, respectively. The He* absorbance indicating the relative n_{He^*} for the liquid anode decreased approximately a half during the discharge pulse, whereas the He* absorbance was stable for the metal anode, according to the results of LAS. The results of the electron-density diagnostics, the LAS, and also the high-speed camera observation suggested the decrease of excited-species densities during the discharge pulse only for the case of liquid anode. The given plasma parameters would be references for following diagnostic studies and basic parameters for simulation studies of the plasmas in contact with liquid.

This study indicated that the water electrode affects not only the generation of species originating from water but also the fundamental plasma parameters even 500 µm above the water surface. The decreases in n_e and n_{He^*} are probably caused by the increase in the amount of water vapor in the discharge space. The increase of water vapor during the discharge pulse was indicated from the results of PMDI given in section 3.3; however, the amount of water vapor could not be evaluated yet. Also, the effects of water should be of a complex nature because cluster ion and radical formation occur under such a humid condition, as indicated in recent studies using mass spectroscopy [77, 78]. In addition, the water vapor inside the discharge region will probably be dissociated and ionized, and these species also affects the reaction kinetics of the plasma. Therefore, it is necessary to reveal the water distribution and develop a way to simulate the reaction kinetics in such a humid condition considering the multiple effects of water and water-related species.

5. Concluding remarks

This paper provides multiple plasma parameters in an atmospheric-pressure pulsed-dc discharge plasma source, which can be operated with a metal or flowing water anode. By the practice using the discharge operated with metal and liquid anode in the same geometry, it was possible to observe the effects of liquid water anode on the plasma parameters in gas phase, as differences from the data obtained with the metal anode. The following three laser diagnostic experiments were conducted: (1) Laser Thomson scattering (LTS) and rotational Raman scattering, (2) Phase-modulated dispersion interferometry (PMDI), and (3) Laser absorption spectroscopy (LAS). We also observed the stability and temporal behavior of the plasma structure using a digital camera and a high-speed video camera.

From the results of LTS, it was revealed that the electron plasma parameters n_e and T_e were approximately 2×10^{14} cm^{-3} and 2 eV, respectively, in the center of the plasma column. The PMDI results indicated n_e in the 10^{14} cm^{-3} order but lower with a factor of two from the LTS results. The results of both LTS and PMDI suggested that n_e decreased by about 20%–35% during a pulse only in the case of the flowing liquid anode. n_{He^*} also decreased by ~50% during a pulse only in the case of the liquid anode, as suggested by LAS. Considering the fact that the ratio of the molecular impurity in the He core flow increased during the discharge pulse only in the case of the liquid anode, as indicated by PMDI, a possible reason for the decreases in n_e and n_{He^*} for the liquid anode is the increase in the amount of water vapor in the discharge space, which evaporated from the surface of the liquid anode. We reached a conclusion from these results that the multiple laser-aided diagnostics used in this study complemented each other not only in the plasma parameter measurement but also in investigation of the water effects on the discharge.

In addition, this study also clarified some topics which are important to be addressed. One is optical emission spectroscopy (OES) diagnostics. The spatiotemporally resolved OES would help to analyze N_2 impurity ratio in the discharge space by providing N_2 vibrational temperature as mentioned in [56] and confirm the increase of water vapor by OH radical emission. Another topic is quantitative measurement of water vapor density in and around the plasma with high spatiotemporal and density resolutions, to diagnose the spatiotemporal change of gas composition during the plasma generation with the liquid electrode. The continuous efforts on diagnostics of plasma and background parameters in gas phase near the liquid are highly expected to contribute to further understandings of the plasmas in contact with liquid.

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